

CBSE Test Paper 03
Chapter 4 Motion in A Plane

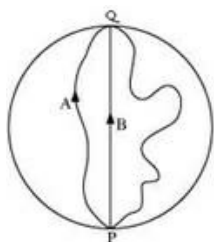
1. A body is projected with a velocity of 20ms^{-1} at 50° to the horizontal. Find Time of flight. **1**
 - a. 4.2 s
 - b. 3.5 s
 - c. 5.1 s
 - d. 3.1 s

 2. If $\mathbf{r} = x\hat{i} + y\hat{j}$ is the position vector of a particle then the instantaneous acceleration is given by **1**
 - a. $a = \frac{dr}{dt}$
 - b. $a = \frac{d^4r}{dt^2}$
 - c. $a = \frac{d^2r}{dt^2}$
 - d. $a = \frac{d^3r}{dt^3}$

 3. The addition of vectors and the multiplication of a vector by a scalar together gives rise to **1**
 - a. distributive laws
 - b. commutative law
 - c. asymmetric laws
 - d. intransitive law

 4. Passengers on a carnival ride move at constant speed in a horizontal circle of radius 5.0 m, making a complete circle in 4.0 s. What is their acceleration? **1**
 - a. 14 m s^{-2}
 - b. 16 m s^{-2}
 - c. 12 m s^{-2}
 - d. 15 m s^{-2}
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5. A body is projected with a velocity of 20ms^{-1} at 50° to the horizontal. Find Maximum height reached. **1**
- $\approx 12\text{m}$
 - $\approx 22\text{m}$
 - $\approx 15\text{m}$
 - $\approx 20\text{m}$
6. Why does the direction of motion of a projectile become horizontal at the highest point of its trajectory? **1**
7. A football is kicked into the air vertically upwards. What is its (i) acceleration and (ii) velocity at the highest point? **1**
8. What are the minimum number of forces which are numerically equal whose vector sum can be zero? **1**
9. Three girls skating on a circular ice ground of radius 200 m start from a point P on the edge of the ground and reach a point Q diametrically opposite to P following different paths as shown in figure. What is the magnitude of the displacement vector for each? For which girl is this equal to the actual length of the path skated? **2**



10. A motor boat is racing towards North at 25 km/h and the water current in that region is 10 km/h in the direction of 60° East to South. Find the resultant velocity of the boat. **2**
11. A particle starts from origin at $t = 0$ with a velocity $15\hat{i}\text{ m/s}$ and moves in xy - plane under the action of a force which produces a constant acceleration of $15\hat{i} + 10\hat{j}\text{ m/s}^2$. Find the y - coordinate of the particle at an instant when its x coordinate is 125 m . **2**
12. The position of a particle is given by $\mathbf{r} = 3.0t\hat{i} + 2.0t^2\hat{j} + 5.0\hat{k}$ where t is in seconds and the coefficients have the proper units for $\mathbf{r}$ to be in metres. **3**

i. Find $v(t)$ and $a(t)$ of the particles.

ii. Find the magnitude and the direction of $v(t)$ at $t = 2.0$ s.

13. A fighter plane flying horizontally at an altitude of 1.5 km with speed 720 km/h passes directly overhead an anti-aircraft gun. At what angle from the vertical should the gun be fired for the shell with muzzle speed 600 m s^{-1} to hit the plane? At what minimum altitude should the pilot fly the plane to avoid being hit? (Take $g = 10 \text{ m s}^{-2}$). **3**

14. A bullet fired at an angle of 30° with the horizontal hits the ground 3.0 km away. By adjusting its angle of projection, can one hope to hit a target 5.0 km away? Assume the muzzle speed to be fixed, and neglect air resistance. **3**

15. A fighter plane is flying horizontally at an altitude of 1.5 km with speed 720 km/h. At what angle of sight (w.r.t horizontal) when the target is seen, should the pilot drop the bomb in order to attack the target? **5**

Main concept used: $u = 720 \text{ km/h} = 720 \times \frac{5}{18} \text{ m/s} = 200 \text{ m/s}$